

# Sophia Zhang

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## EDUCATION

### Massachusetts Institute of Technology

Cambridge, MA

*B.S. in Computer Science and Mathematics*

*Aug. 2025 – May 2028 (expected)*

- Coursework: Inference & Information (G), Probability (G), Machine Learning (G), TinyML and Efficient AI Computing (G), Operating Systems, Computation Structures

### Princeton International School of Mathematics and Science (PRISMS)

Princeton, NJ

*High school*

*Sep. 2022 – June 2025*

## EXPERIENCE

### Undergraduate Researcher (UROP)

Sep. 2025 – Present

*Dina Katabi's Lab, MIT CSAIL*

*Cambridge, MA*

- Working with diffusion models, multimodal foundation models, and recommendation models.

### AI Engineering Intern

June – Aug. 2026

*Alpha School*

*Austin, TX*

- Built RL (GRPO) training setups and evaluations for LLM tutors and educational apps, grounded in learning science and pedagogy.

### Trading & Technology Intern

Jan. 2026

*Hudson River Trading*

*New York, NY*

- Software engineering and quant research projects, including trading Brazilian equities.

### Research Collaborator – LLMs for Formal Theorem Proving

June 2025 – Jan. 2026

*Microsoft Research Asia*

*Remote*

- Verified Lean formalizations of math proofs and contributed to system improvements; two papers below.

## PUBLICATIONS

STAR-PólyaMath: Multi-Agent Reasoning under Persistent Meta-Strategic Supervision

NeurIPS 2026 (Spotlight)

- Jiaao Wu, Xian Zhang, Hanzhang Liu, **Sophia Zhang**, Fan Yang, Yinpeng Dong

Euclean: Automated Geometry Problem Formalization with Unified Verification in Lean

ICML 2026

- Linbin Tang, Jingyan You, Zilin Kang, Hanzhang Liu, **Sophia Zhang**, Zenan Li, Chenrui Cao, Liangcheng Song, Jiaao Wu, Xian Zhang, Fan Yang

## PROJECTS

AI Math Olympiad Progress Prize 3 | *Python, PyTorch, vLLM, Unsloth*

Jan. – Apr. 2026

- Competed with a team of MIT freshmen, all Olympiad medalists. Built for the training and inference scaffolding (vLLM, Unsloth), hidden-state extraction, and parameter sweeps.
- Authored the team's technical writeup.

## AWARDS

**Putnam** Top 200, 2026

**USAMO** Silver, 2025

**USAPhO** Honorable Mention, 2025

**Mathematical Olympiad Program (MOP)** Two-time attendee, 2023, 2024

## TECHNICAL SKILLS

**Languages:** Python, C++, C, Rust, Java, TypeScript, Lean, RISC-V Assembly, Minispec, Dafny

**AI Research:** RL post-training, diffusion models, multimodal models, LLM agents & multi-agent reasoning, formal theorem proving, recommendation models

**Tools:** PyTorch, vLLM, Unsloth, NumPy, SciPy, Pandas, Matplotlib, Jupyter, automated eval pipelines, AI coding agents & multi-agent workflows